

Swiss house-type classification

MACHINE LEARNING PROJECT / 06-DUMAU10 2021/SL

Objective

Predict the type of a Swiss residential property from its price, floor area, number of rooms, and location.

Data

The original report records 16,701 examples: 13,361 for training and 3,340 for testing. Missing numerical values were replaced with the mean; missing categorical values were replaced with the most frequent value.

Source: [Switzerland House Price Prediction Data](#), Etienne Kaiser / Kaggle.

Models described in the original report

Logistic regression, with and without L2 regularization. The report describes third-degree polynomial features and mini-batch gradient descent with batches of 100.

Gaussian naive Bayes. Class-conditional features were modeled with normal distributions.

Neural network. One hidden layer with 32 ReLU units, a softmax output layer, and the Adam optimizer.

Recorded results

Model	Accuracy	Precision	Recall	F1 score
Logistic regression + L2	0.5862	0.4092	0.5862	0.4625
Logistic regression, no regularization	0.5877	0.4331	0.5877	0.4866
Gaussian naive Bayes	0.3051	0.5529	0.3051	0.3737
Neural network	0.6686	0.7140	0.6686	0.6007

Conclusions

The neural network achieved the highest recorded values for all four metrics. The two logistic regression models performed similarly. Naive Bayes had lower accuracy but higher precision than either logistic regression model.

Reproducibility

These values are translated historical results, not a new benchmark. The archived script used standard scikit-learn logistic regression without polynomial feature expansion or the stated mini-batch optimizer. The refactored script retains the implemented model families and now fits preprocessing only on training data. The source CSV is not included, so these scores have not been reproduced. An 80/20 scikit-learn split of 16,701 rows produces 13,360 training and 3,341 test rows, differing from the counts recorded above.